A RETROWAVE TRANSMISSION

THE FUTURE OF

AI + BLOCKCHAIN

A JOURNEY THROUGH TIME · PROTOCOLS · AGENTS

FROM DIGITAL LEDGERS → AUTONOMOUS ECONOMIES

▷ 2026 EDITION

TRANSMISSION LOG

A FIVE-PART JOURNEY

I HISTORICAL RETROSPECTIVE

Parallel origins — from Turing to Bitcoin to the Transformer

II THE CURRENT STATE

Tokenized compute, agentic wallets, sentient protocols

III THE PROTOCOL STACK

MCP · A2A · x402 · AP2 · MPP — rails of the agent economy

IV AGENTIC PAYMENTS BOT

A deep dive into Sentient-AGI's dual-protocol payment skill

V THE FUTURE FINANCIAL SYSTEM

Machine economies, pay-per-inference markets, autonomous capital

PART I

HISTORICAL RETROSPECTIVE

▷ 1940s → 2024

TWO PARALLEL DREAMS

THE ORIGINS — 1940s → 2008

🧠 ARTIFICIAL INTELLIGENCE

- 1943 — McCulloch & Pitts: first mathematical neuron.
- 1950 — Turing proposes the Imitation Game.
- 1956 — Dartmouth workshop coins "AI".
- 1969 — Minsky & Papert kill the perceptron; first AI winter.
- 1986 — Backpropagation revived (Rumelhart, Hinton, Williams).
- 1997 — Deep Blue beats Kasparov; LSTM is born.
- 2006 — Hinton's deep belief nets reignite the field.

⛓ CRYPTO & LEDGERS

- 1976 — Diffie–Hellman: public-key cryptography.
- 1983 — David Chaum proposes eCash / blind signatures.
- 1991 — Haber & Stornetta: cryptographic time-stamping.
- 1997 — Adam Back: Hashcash proof-of-work.
- 1998 — Wei Dai's b-money, Nick Szabo's bit gold.
- 2004 — Hal Finney's Reusable Proof-of-Work.
- 2008 — Satoshi's Bitcoin whitepaper drops.

Two disciplines — one computational, one monetary — circling the same question: can a machine reason, and can a machine remember?

PARALLEL EVOLUTION

DEEP LEARNING × DECENTRALIZED MONEY — 2009 → 2016

2009
BITCOIN GENESIS

Block 0 mined — first trustless digital scarcity goes live.

2012
GPU-TRAINED VISION

AlexNet crushes ImageNet — deep learning hits industrial scale.

2013
ETHEREUM PROPOSED

Vitalik: 'a blockchain with a Turing-complete programming language'.

2014
GANs + WORD2VEC

Generative models go mainstream; language becomes geometry.

2015
ETHEREUM MAINNET

Smart contracts ship — programmable money lands.

2016
ALPHAGO + THE DAO

AI beats Lee Sedol; DeFi's first catastrophic hack teaches humility.

Two revolutions — running in parallel, rarely speaking — each builds primitives the other will soon need.

THE CONVERGENCE

WHEN PATHS FIRST CROSSED — 2017 → 2021

AI SIDE

- 2017 — 'Attention Is All You Need' introduces the Transformer.
- 2018 — BERT + GPT-1: pretrained language models arrive.
- 2019 — GPT-2 withheld, then released — scaling is real.
- 2020 — GPT-3 (175B) turns few-shot learning into a product.
- 2020 — AlphaFold 2 solves 50-year-old protein problem.
- 2021 — DALL·E, Codex: generative AI crosses modalities.

BLOCKCHAIN SIDE

- 2017 — ICO boom; ERC-20 turns any idea into a token.
- 2018 — Layer-2 research begins (Plasma, state channels).
- 2020 — DeFi summer: Uniswap, Compound, Aave hit scale.
- 2020 — USDC launches — stablecoin rails solidify.
- 2021 — NFT explosion; on-chain identity becomes real.
- 2021 — Solana, rollups — the speed problem gets solved.

CONVERGENCE SIGNAL → programmable money needs programmable intelligence — and vice versa.

THE AGENT ERA BEGINS

WHEN MODELS STARTED DOING THINGS — 2022 → 2024

NOV 2022

CHATGPT

100M users in 60 days — the fastest product adoption in history.

2023

CLAUDE / GPT-4 / LLAMA

Foundation models commoditize; open-weights race begins.

2023

AGENTIC FRAMEWORKS

AutoGPT, BabyAGI, LangChain — LLMs become loops, not oracles.

2023

TOKENIZED RWAs

Treasuries, real estate, commodities move on-chain at scale.

NOV 2024

MCP LAUNCHES

Anthropic ships Model Context Protocol — tool use standardized.

2024

CRYPTO-AI CONVERGENCE

Sentient, Bittensor, Fetch — AI networks settle on blockchain.

By the end of 2024, AI models are agents. Blockchain rails are real. The only missing piece: a way for them to pay each other.

PART II

THE CURRENT STATE

▷ 2025 // 2026

THE STATE OF THE UNION

WHERE AI AND BLOCKCHAIN STAND — 2026

$600M

x402 annualized volume

119M tx on Base · 35M on Solana

97M

MCP SDK downloads / mo

From 100K in Nov 2024 → 97M in 2025

60+

AP2 protocol partners

Mastercard · Amex · PayPal · Coinbase

AGENTIC WALLETS

AI agents now hold & control their own keys — via KMS (Turnkey, AWS), embedded wallets (Privy), or smart accounts (EIP-4337).

TOKENIZED COMPUTE

GPU time and model inference trade on-chain — Bittensor, Akash, io.net, Render settle in their own tokens.

STANDARDS CONSOLIDATION

Linux Foundation AI Agent governance. MCP + A2A unified. Visa & Mastercard join the agentic stack.

CASE STUDY : SENTIENT PROTOCOL

LOYAL AI × ON-CHAIN OWNERSHIP — a blueprint for the open AGI economy

THE PROBLEM IT SOLVES

Closed LLMs own the value chain
OpenAI, Anthropic, Google capture revenue. Open-source contributors get nothing.

No ownership, no provenance
Once weights leak, there's no way to trace usage or pay authors.

No incentive to compose or improve
Fine-tuning a base model = free labor for the original lab.

THE OML FRAMEWORK

O — OPEN
Weights are public. Anyone can deploy, audit, or fine-tune.

M — MONETIZABLE
Model fingerprints + on-chain ledger track every inference call; smart contracts distribute fees to contributors.

L — LOYAL
Models respond only to authorized callers. TEE attestation (AWS Nitro Enclaves) ensures no silent theft.

Four protocol layers · Storage → Distribution → Access → Incentive (smart contracts route every cent)

PART III

THE PROTOCOL STACK

▷ MCP · A2A · x402 · AP2 · MPP

MCP

MODEL CONTEXT PROTOCOL

Anthropic · Nov 2024 · the data / context layer

⚠ SEPARATE FROM PAYMENTS — MCP is the connective tissue for AI ↔ data and tools, not for value transfer.

WHAT IT IS

"USB-C for AI applications"

A client–server protocol that lets any LLM securely connect to any external data source, tool, API, or other AI system — without custom integrations.

Host (Claude, Cursor, IDE)
Client (protocol handler)
Server (exposes Tools · Resources · Prompts)

WHY IT MATTERS

Solves the M × N problem
N models × M tools no longer = N×M custom glue. One standard, many servers.

Vertical integration layer
Agent sits on top, tools underneath. MCP makes the vertical connection.

Governance
Donated to Linux Foundation's Agentic AI Foundation alongside A2A.

DATA + CONTEXT — that's MCP's lane. Payments live on the stack above.

A2A

AGENT - TO - AGENT PROTOCOL

Google · April 2025 · the horizontal / collaboration layer

AGENT CARDS

JSON manifest at /.well-known/agent.json — an agent's digital business card: skills, endpoint, auth.

TASKS

Every interaction is a stateful Task object with a lifecycle — from quick lookups to multi-day workflows.

MESSAGES & PARTS

Multipart messages carry text, JSON, files, even generated images across the agent mesh.

TRANSPORTS

Built on HTTP + SSE + JSON-RPC — OAuth 2.0, PKCE, API keys out of the box. No reinventing.

MCP ≠ A2A

MCP connects one agent DOWN to tools and data. A2A connects agents ACROSS to each other. Google: "A2A complements MCP."

x402

HTTP - NATIVE STABLECOIN PAYMENTS

Coinbase · May 2025 · reviving the dormant HTTP 402 code

THE HANDSHAKE / FIVE STEPS

01 REQUEST

Client calls a gated endpoint (agent or human).

02 402

Server replies 402 Payment Required + PAYMENT-REQUIRED header.

03 SIGN

Client constructs a payment payload, signs with its wallet.

04 VERIFY

Facilitator verifies on-chain; EIP-3009 (USDC) or Permit2.

05 DELIVER

Server returns 200 + resource + PAYMENT-RESPONSE receipt header.

$0.0001

per-tx settlement cost

< 2 sec

on-chain finality

ZERO

protocol fees (only gas)

5+

networks: Base · ETH · Polygon · Arbitrum · Solana

Built on the Linux/Cloudflare x402 Foundation · supported by AWS Bedrock, Cloudflare Workers, Vercel, Google

AP2

AGENT PAYMENTS PROTOCOL

Google + 60 partners · Sep 2025 · the trust / authorization layer

THREE CRYPTOGRAPHICALLY SIGNED MANDATES

01

INTENT MANDATE

"Buy running shoes under $100."

Captures the user's initial intent — spending ceiling, timing, and constraints. Signed by the user upfront.

02

CART MANDATE

Locks specific items + final price.

After the agent finds a match, the merchant cryptographically signs the cart — a guarantee of fulfillment.

03

PAYMENT MANDATE

Authorizes actual settlement.

Shared with the payment network. Signals agent-presence to help issuers assess risk and route rails.

PAYMENT-AGNOSTIC cards · bank rails · stablecoins (via A2A x402 extension)

ACCOUNTABILITY non-repudiable chain of cryptographic evidence: intent → cart → payment

BUILT ON A2A for transport · MCP for context · compatible with Visa, Mastercard, PayPal flows

MPP

MACHINE PAYMENTS PROTOCOL

Stripe + Tempo Labs · March 2026 · IETF Standards Track draft

◆ *First payment protocol designed for an entity with no psychology — no checkout, no cart abandonment, no mental transaction cost.*

SESSIONS

Agent pre-authorizes a spending budget once. Then streams thousands of signed micropayments off-chain → one on-chain settlement.

METHOD-AGNOSTIC

Stablecoins (Tempo L1), fiat via Stripe's Shared Payment Tokens, Bitcoin Lightning, Visa cards — all via one HTTP interface.

NO FACILITATOR

Architecturally eliminates the x402 facilitator. Payment verification is a first-class primitive of the protocol itself.

BACKWARD-COMPAT

Backward-compatible with x402's charge flow. MPP clients can consume existing x402 services unchanged.

Day-one adopters: OpenAI · Anthropic · Google Gemini · Dune · Browserbase · Parallel · Visa · Mastercard · Lightspark → 100+ listed services

MPP sits at the settlement + session layer. AP2 is the trust layer above. MCP is the context layer beside it.

THE FULL STACK

HOW THE FIVE PROTOCOLS COMPOSE

L5 AP2 — **INTENT & TRUST**
Mandates · verifiable credentials · accountability chain

L4 A2A — **AGENT COLLABORATION**
Agent cards · tasks · multi-part messages across vendors

L3 MPP — **SETTLEMENT & SESSIONS**
Streaming micropayments · fiat + stablecoin via one API

L2 x402 — **PAYMENT HANDSHAKE**
HTTP 402 · on-chain USDC in sub-2-second finality

L1 MCP — **CONTEXT & TOOLS**
Data access · tool invocation · per-model context plumbing

Context flows up. Money flows down. Agents plug into both ends.

PART IV

AGENTIC PAYMENTS BOT

▷ SENTIENT-AGI · github.com/sentient-agi/agentic-payments-bot

AGENT - PAYMENTS - SKILL

A DUAL-PROTOCOL PAYMENT SKILL FOR THE OPEN AGENT SKILLS ECOSYSTEM

A dual-protocol (x402 + AP2) agentic payment service enabling AI agents to autonomously initiate, validate, and execute payments across blockchain (web3) and traditional (web2) rails — with AWS KMS key management, a policy engine, audit trail, and human-in-the-loop confirmation.

DUAL PROTOCOL

Native x402 (web3) + AP2 (web2/cards) in one skill

AWS KMS

Keys never leave the hardware module; ECDSA signing via KMS

POLICY ENGINE

Per-tx limits · daily/weekly/monthly aggregates · blacklists · time windows

MULTI-GATEWAY

Viem for EVM · Stripe · PayPal · Visa · Mastercard — one router

AUDIT TRAIL

SQLite-backed immutable log · every state transition captured

HUMAN-IN-THE-LOOP

Configurable confirmation thresholds · agent alone or with approval

Compatible clients: Claude Code · Codex · Junie · OpenCode · GitHub Copilot · Gemini CLI · OpenClaw

THE PIPELINE

FROM AGENT INTENT TO SETTLED PAYMENT

1 AGENT OUTPUT
LLM emits paymentIntent JSON

2 ZOD VALIDATE
Schema + type safety

3 PROTOCOL ROUTER
x402 if web3 · AP2 if web2

4 POLICY ENGINE
Limits · time · blacklist · YAML

5 HITL GATE
Threshold? → await human approval

6 KMS SIGN
AWS KMS ECDSA signature

7 EXECUTE
Viem / Stripe / PayPal settle

TRANSACTION STATE MACHINE

`pending` → `policy_check` → `awaiting_confirmation` → `approved` → `executed`

↳ `rejected`

↳ `failed`

SQLite · transactions (id · protocol · gateway · amount · status · tx_hash · violations)

SQLite · audit_log (timestamp · level · category · action · actor · details)

HOW THE TWO RAILS WORK

INSIDE THE DUAL-PROTOCOL ENGINE

x402 · WEB3 RAIL

Detect 402
Parse PAYMENT-REQUIRED header to find accepted networks + tokens.

Build Payment Payload
Choose scheme (exact, deferred). Currently: Base, Polygon, Arbitrum, Solana.

Sign via KMS
AWS KMS ECDSA — private key never in memory. Viem handles broadcast.

Retry with PAYMENT-SIGNATURE
Facilitator verifies, server returns resource + receipt header.

AP2 · WEB2 RAIL

Create Mandate
From the PaymentIntent build an AP2 Intent Mandate: amount · window · delegator.

Sign Mandate
Sent to a credential provider — ECDSA-based verifiable credential signed by user.

Obtain Credentials
Payment method creds fetched for the chosen gateway (Stripe · PayPal · Visa · MC).

Execute Charge
Gateway adapter pulls funds, returns tx_id + status, all logged.

POLICY ENGINE runs BEFORE both flows — no signature is produced if a rule violates, and violations are recorded in the audit log.

PART V

THE FUTURE FINANCIAL SYSTEM

▷ A MACHINE ECONOMY AT THE SPEED OF LIGHT

THE MACHINE ECONOMY

WHAT HAPPENS WHEN AGENTS CAN TRANSACT WITHOUT US

FROM SUBSCRIPTIONS → PAY-PER-TOKEN

Agents don't tolerate flat fees. APIs price by the kilobyte, the query, the inference. Monthly tiers evaporate.

FROM CHECKOUT CARTS → STREAMING PAYMENTS

MPP sessions authorize a budget once and pay as the agent consumes — no cart, no click, no abandonment.

FROM KYC → KNOW YOUR AGENT

Agent identity (JWT · ERC-8004 · Skyfire KYA) replaces account signup. Verified agents get paid. Bots get blocked.

FROM CAPITAL ALLOCATION → AUTONOMOUS TREASURY

Agents holding their own wallets hire other agents, buy compute, rebalance portfolios — all under policy guardrails.

Rebuilt on two primitives: cryptographic intent + programmable settlement.

NEW ECONOMIC PARADIGMS

MODELS THAT WEREN'T POSSIBLE BEFORE

INFERENCE MARKETS

Open-weight models bid for your query. Router picks cheapest that meets your spec. Paid per-token, on-chain.

DATA AS AN ASSET

Training data sold per row. Provenance traced on-chain. Contributors paid continuously as models train on their work.

COMPUTE FUTURES

GPU hours tokenized and traded. Prices discover themselves. Agents hedge inference costs like airlines hedge fuel.

AGENT-TO-AGENT DeFi

Agents provide liquidity to other agents. Lending, collateral, and swaps happen entirely among machine counterparties.

MICRO-COMMERCE

$0.0001 reads become viable. Every paragraph paywalled. Every crawl monetized. Every API call its own micro-economy.

DELEGATED CAPITAL

User signs: "rebalance my portfolio, keep ≥$5k USDC, don't exceed 10% risk". Agent executes for months unattended.

Each of these requires: identity + policy + payment + settlement — the full protocol stack we just walked through.

END OF TRANSMISSION

THE HORIZON

AI + Blockchain is no longer two fields.

It is one substrate for machine-native commerce — already live, already settling.

CONTEXT	TRUST	SETTLE
MCP · A2A	*AP2 mandates*	*x402 · MPP*

The agents are here. The rails are live. The economy is bootstrapping itself.

THANK YOU